

Spring 2026 Midterm 1

The exam is closed book and closed notes. You are allotted one 8.5" x 11", double sided, "cheat sheet" of handwritten notes. You may also use a simple scientific (non-graphing) calculator. Your phone is not allowed as a calculator. Using any resources outside of the aforementioned items is strictly prohibited.

While you take the exam, you are prohibited from discussing the test with anyone. If you are taking the test after your classmates, you are also prohibited from talking to them about the test before you take it. Evidence of cheating may result in a 0 on the exam and be reported to the Student Conduct Board.

First and Last Name (print clearly): _____

Student ID Number (print clearly): _____

Please read the following statement and sign below to indicate that you understand the policies for this exam
You must show your student ID when you submit your test.

UC Berkeley Honor Code

"As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others." Please carefully read the statements below, and indicate your understanding and intent to adhere to the UC Berkeley Honor code by signing your name in the space below. I agree not to engage in any of the following behaviors:

- Copying or attempting to copy from others during an exam or on an assignment.
- Communicating answers with another person during an exam.
- Pre-programming a calculator or other personal electronic device to contain answers, or using other unauthorized information for exams.
- Using unauthorized materials, i.e. prepared answers.
- Allowing others, or utilizing software like generative AI, to do an assignment or a portion of an assignment for you, including the use of a commercial term-paper service.
- Submitting the same assignment for more than one course without prior approval of all the instructors involved.
- Collaborating on an exam or assignment with any other person without prior approval from an instructor.
- Taking an exam for another person or having someone take an exam for you.
- Altering a previously graded exam or assignment for the purpose of a grade appeal or of gaining points in a re-grading process.
- Submitting an electronic file the student knows to be unreadable or corrupted instead of a completed assignment.

Signature:

Sign your name:

INSTRUCTIONS:

Handwrite your responses using a pencil or pen in the space provided. Give your responses **ONLY** in the space provided. Do not write your responses outside of the provided spaces. The additional space provided, including space on the back side of the exam can be used as scratch paper but will not be graded.

Phones should be turned OFF prior to the start of the exam and secured in your backpack or another secure location. Do not leave your phone or other electronic devices out. If you need to leave the room for any reason during the exam please flag a GSI and let them know prior to exiting the room. Upon leaving the exam room, you must leave your electronic device(s) with the GSIs. Time will still accrue when you leave the room.

The length of the midterm is 50 minutes. If you finish early and are satisfied with your work you may leave early. Hand in your exam to a GSI, who will verify that they received it, with your student ID.

- Unless otherwise specified in the question, format your answers according to the following guidelines:
 - present your answers rounded to two decimal places
 - present proportions as decimal values rounded to 2 decimal places rather than percents (ex: 0.41 rather than 41.30%)
- All logarithms (“logs”) are the natural logarithm (base e ; $\log_e(x)$; $\ln(x)$)

Exam Format:

Short Format Questions: Q1-6 [15 points]

Long Format Question: Q7 [10 points]

Long Format Question: Q8 [10 points]

Long Format Question: Q9 [10 points]

Extra credit: Q10 [2 points] Optional Feedback Question

Short Format Questions: Please select only one answer choice.

1. [2 points] A county health department collects data from a random sample of residents to estimate the proportion who have been diagnosed with hypertension. What type of problem is being addressed?

- ☐ A. Descriptive
- ☐ B. Predictive
- ☐ C. Causative/Etiologic
- ☐ D. Exploratory

2. [2 points] Based on the following table containing pet category totals, which ggplot code will correctly plot count as bar heights?

```
##   animal count
## 1    Dog    12
## 2    Cat     9
## 3   Fish     4
## 4  Horse     1
## 5 Lizard     1
```

- ☐ A. `ggplot(pets, aes(x = animal)) + geom_bar()`
- ☐ B. `ggplot(pets, aes(x = animal, y = count)) + geom_bar(stat = "identity")`
- ☐ C. `ggplot(pets, aes(x = count)) + geom_histogram()`
- ☐ D. `ggplot(pets, aes(x = animal, fill = count)) + geom_bar()`

3. [2 points] A simple linear regression finds a strong positive relationship between number of coffee drinks per day and annual income. Which interpretation is most appropriate?

- ☐ A. The regression shows that coffee consumption causes income to increase
- ☐ B. The regression shows that income and coffee consumption are associated
- ☐ C. The regression shows that earning a higher income causes more coffee consumption
- ☐ D. The regression result cannot be interpreted at all

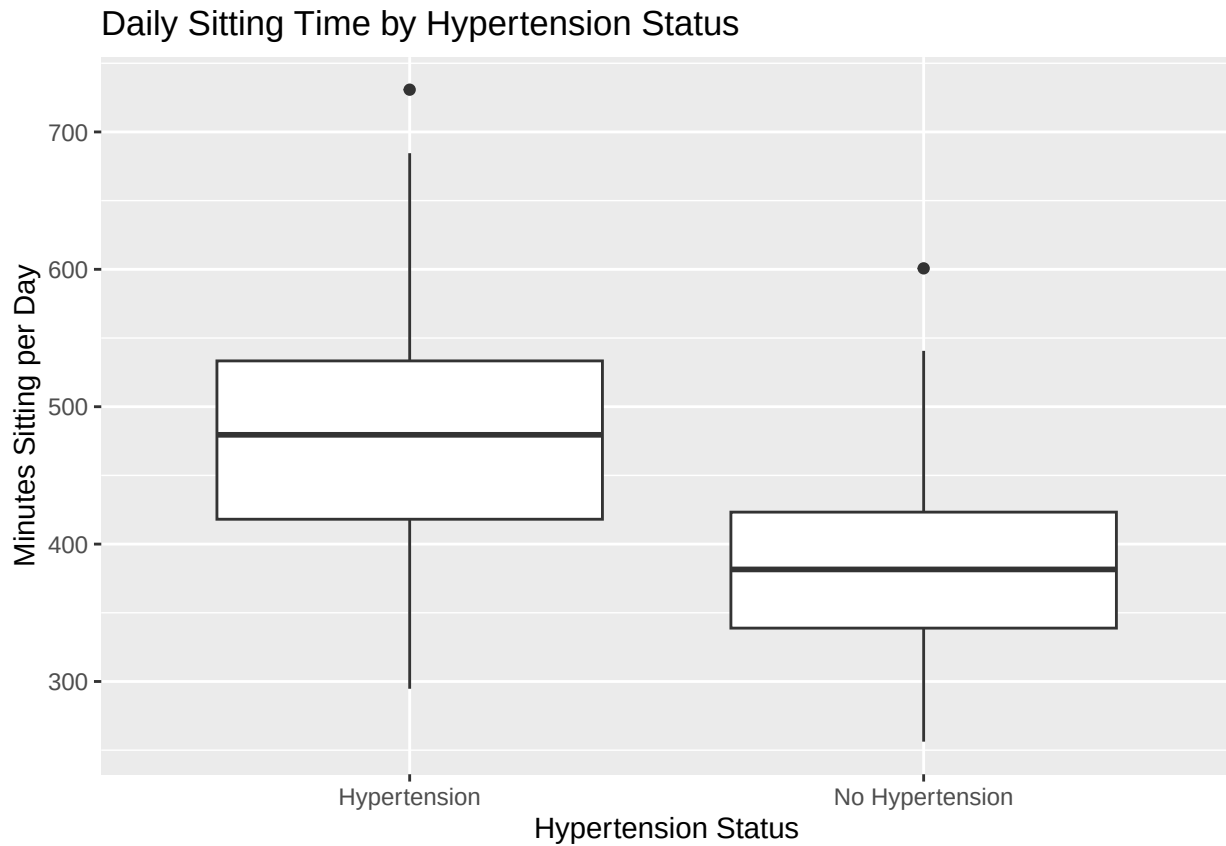
4. [2 points] A researcher fits the following regression model relating house size (square feet) to monthly electricity cost:

$$\log(\text{Cost}) = 4.20 + 0.0005 \times \text{Size}$$

What is the monthly electricity cost (in dollars) for a 2,000 square-foot house?

- ☐ A. 52
- ☐ B. 320
- ☐ C. 520
- ☐ D. 181

5. [2 points] Based on the boxplot below, which of the following statements is TRUE? (Select one)



- ☐ A. Individuals with hypertension have a lower median sitting time than those without hypertension.
- ☐ B. The variability (IQR) in sitting time appears larger among individuals with hypertension than among those without hypertension.
- ☐ C. Both groups have identical distributions of sitting time.
- ☐ D. The boxplot shows that sitting longer causes hypertension.

Q6 [5 points]

Read the following abstract and answer the questions below.

Researchers conducted a 24-week randomized clinical trial to evaluate whether a weekly injectable GLP-1 receptor agonist leads to greater weight loss than placebo among adults with obesity ($\text{BMI} \geq 30 \text{ kg/m}^2$). Participants were recruited through advertisements posted at a private fitness club and a high-end medical spa in San Francisco. A total of 180 volunteers enrolled and were randomly assigned to receive either the GLP-1 medication or placebo. Body weight (kg) was measured at baseline and week 24. The primary outcome was absolute change in body weight (kg) from baseline to week 24.

6.1 [1.5 points] What is the exposure variable?

- ☐ A. Body mass index (BMI)
- ☐ B. GLP-1 medication vs placebo
- ☐ C. Body weight (kg)
- ☐ D. Recruitment location

6.2 [1.5 points] What kind of variable is the outcome?

- ☐ A. Nominal
- ☐ B. Ordinal
- ☐ C. Discrete
- ☐ D. Continuous

6.3 [2 points] Which issue most threatens the generalizability of this study's findings?

- ☐ A. Sampling bias
- ☐ B. Informed consent
- ☐ C. Adherence
- ☐ D. Equipoise

Long Format Questions

Q7 [10 points]

Read the following abstract and answer the questions below.

Researchers at UCSF Medical Center conducted a randomized, double-blind clinical trial to evaluate whether a new oral medication (“Drug X”) is more effective than a placebo at reducing fasting blood glucose levels in patients with Type 2 diabetes. A total of 500 patients diagnosed with Type 2 diabetes were recruited from the UCSF Endocrinology Clinic between March and September 2024. Patients were randomly assigned to receive either Drug X (n = 250) or a placebo (n = 250) daily for 16 weeks. Informed consent was obtained from all participants prior to enrollment. The primary outcome was the change in fasting blood glucose (mg/dL) from baseline to week 16. Patients in the Drug X group experienced a significantly greater reduction in fasting blood glucose (mean change: -28.4 mg/dL) compared to the placebo group (mean change: -8.2 mg/dL).

7.1 [2 points] What type of study is this?

- ☐ A. Cross-sectional observational study
- ☐ B. Prospective cohort study
- ☐ C. Randomized controlled experiment
- ☐ D. Case-control study

7.2 [4 points] Can the results of this study be generalized to all adults with Type 2 diabetes in the United States? Explain why or why not (1-2 sentences)

7.3 [4 points] Is confounding likely to explain the observed difference in blood glucose reduction between the Drug X and placebo groups? Explain why or why not (1-2 sentences).

Q8 [10 points]

The following table summarizes smoking status and lung cancer status in a population of 1000 people.

Group	Lung Cancer	No Lung Cancer	Total
Smoker	12	238	250
Non-smoker	7	743	750
Total	19	981	1000

8.1 [2 points] Is this an observational or experimental study?

- ☐ A. Observational
☐ B. Experimental

8.2 [4 points] What is the marginal probability of having lung cancer? Present your final answer as a decimal value (not percent) rounded to two decimal places. Show your work.

Final Answer (round to 2 decimals):

8.3 [4 points] What is the conditional probability of having lung cancer given smoking? Present your final answer as a decimal value (not percent) rounded to two decimal places. Show your work.

Final Answer (round to 2 decimals):

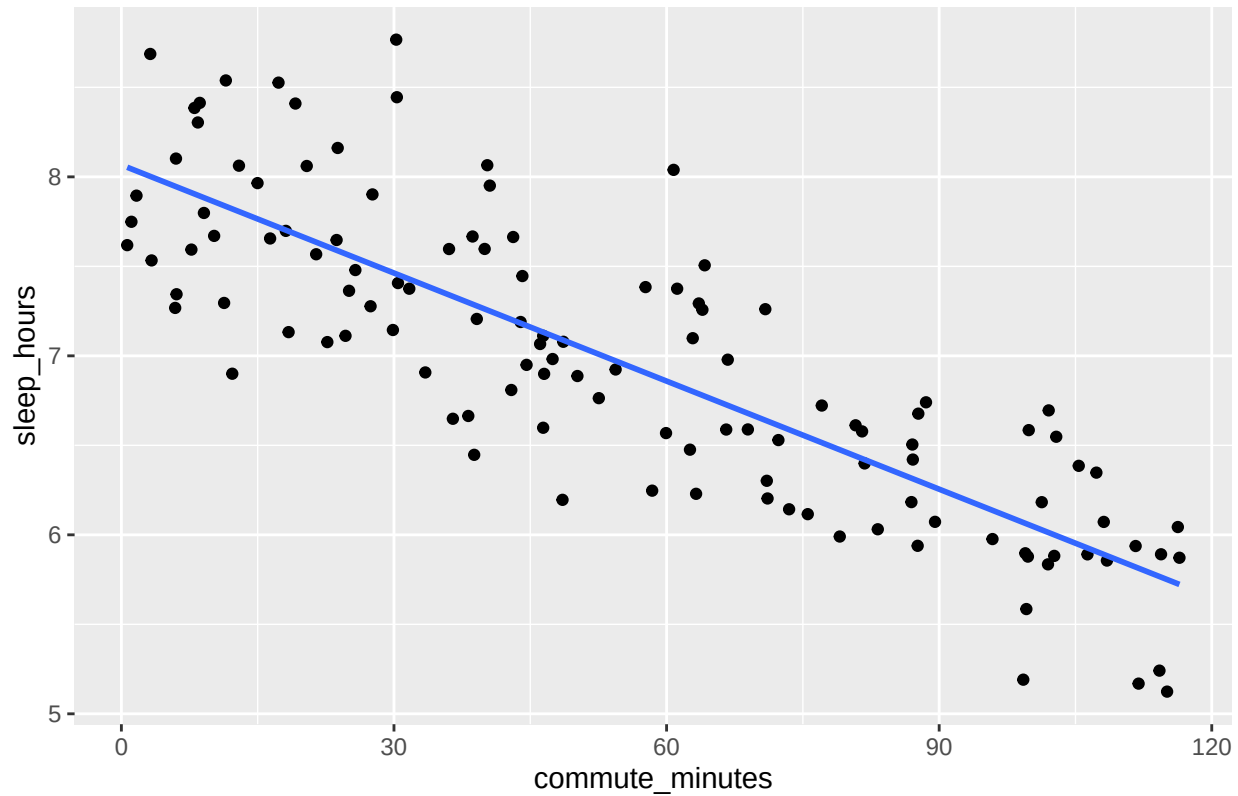
Q9 [10 points]

The scatterplot below shows the relationship between daily commute time (minutes) and hours of sleep per night among 120 Bay Area adults.

A simple linear regression model was fit:

$$\text{sleep_minutes} = 8 - 0.02 \times \text{commute_minutes}$$

Commute Time and Sleep Duration



9.1 [4 points] Fill in the code below that produces the scatter plot above.

```
commute_plot <- ___1___(reg_data, ___2___(x = ___3___, y = sleep_hours)) +  
  ___4___ +  
  geom_smooth(method = "lm", se = FALSE)
```

- 1: _____
- 2: _____
- 3: _____
- 4: _____

9.2 [3 points] Interpret the slope *in context* .

9.3 [3 points] Interpret the intercept *in context* . Is it meaningful? Explain why or why not (1-2 sentences).

10.[2 points] Extra Credit: In one sentence, describe a core principle of the Nuremberg Code

Optional feedback

Take a moment to let us know any issues that came up on the exam.

This space is intended for scratch paper, we will NOT be grading this page